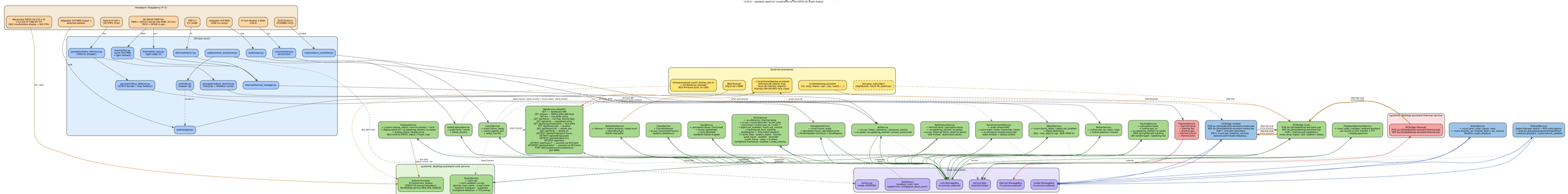

VERA — Vision-Enabled Reasoning Agent — System Architecture

v1.61.0 — playback spectrum visualization on the ESP32-C6 mouth display

Hardware (Raspberry Pi 5)
Waveshare ESP32-C6-LCD-1.47
172×320 ST7789 SPI TFT
(BLE mouth/status display + BLE OTA)
reSpeaker XVF3800 output → attached speaker
Hailo-8 AI HAT+ (26 TOPS, PCIe)
NF-A6x25 PWM Fan
PWM ← GPIO13 (kernel HW PWM, 25 kHz)
TACH → GPIO6 (2 ppr)
TMP117 (I²C 0x48)
reSpeaker XVF3800 (USB mic array)
Pi Cam Module 3 Wide (CSI-0)
DS3218 Servo (PCA9685 ch15)
PCIe
PWM
tach
I²C
USB
CSI
I²C/PWM
SPI + GPIO
USB
Drivers (src/)
perception/hailo_inference.py (VDevice wrapper)
thermal/fan.py (sysfs HW PWM + lgpio fallback)
thermal/fan_tach.py (lgpio edge cb)
thermal/tmp117.py
audio/version_announcer.py
audio/input.py
vision/camera.py (picamera2)
motion/servo_controller.py
perception/face_detector.py (SCRFD decode + Haar fallback)
audio/tts.py (espeak-ng)
perception/object_detector.py (YOLOv8s + letterbox cache)
thermal/thermal_manager.py
renders to
audio/output.py
External processes
firmware/upload_esp32_display_ble.sh
+ nimbleota.py uploader
(BLE firmware push, no USB)
Web Browser http://<pi>:8080
~/.local/share/desktop-assistant/
telemetry.db (SQLite ring)
faces.db (identity registry)
thumbs/ (64×64 JPEG face crops)
scripts/desktop-assistant
(CLI: ping / status / pan / say / watch / ...)
3rd-party subscribers
(dashboards, future ML pipelines)
latest_frame()
latest_chunk() + chunk_index()
latest_frame()
HTTP/WS :8080
IPCClient RPC
(music/podcast reads+actions)
writes telemetry.db
writes faces.db
ZMQ REQ/SUB
ZMQ SUB
ZMQ PUB→SUB (cross-process)
BLE GATT (OTA)
DisplayService
← system.startup_status / service.started / *.error
← display.spectrum / av.speaking_started / av.spoke
→ display.status / display.error
(BLE write to ESP32: status / mouth / eq)
AudioCaptureService
→ audio.level / chunk
+ latest_chunk()
VisionService
→ vision.frame_ready
← vision.capture_still
+ latest_frame()
latest_frame()
WebService (FastAPI)
GET / — dashboard SPA
GET /stream — MJPEG 640×360 feed
WS /ws — live JSON status
GET /api/faces — list (has_thumb flag)
GET /api/skills — registered skills
GET /api/faces/{id}/thumb — face JPEG
PUT /api/faces/{id} — rename
DEL /api/faces/{id} — delete one
DEL /api/faces — delete all
GET|PUT /api/settings/quiet-hours
GET|PUT /api/settings/audio
GET|PUT /api/settings/voice
GET|PUT /api/room
GET|PUT /api/music/* — proxied via IPCClient
GET|POST /api/podcasts/* — proxied via IPCClient
POST /api/say|pan|version|utterance
port 8080
TelemetryService
← thermal.* / motion.position / audio.level
→ telemetry.flush
SQLite ring buffer
ClockService
+ ClockAnnouncer
→ av.say (hourly/half-hourly)
respects QuietHours
FaceService
← perception.faces / face.meet
→ av.say (greetings)
→ face.identified
cooldown-based re-greet
SkillsService
← av.utterance / thermal.temp
→ av.say / vision.describe / av.tell_joke
→ face.meet / motion.pan_to / music.*
→ object.set_enabled / music.set_volume
→ tracking.set_face_tracking
17 skills: help · system_status · volume
quiet_hours · weather · reminder
smart_home · news + 9 others
SkillRegistry — first-match dispatch
ConfigField framework: enabled / config_schema
PerceptionService
← vision.frame_ready
→ perception.faces / perception.error
+ FaceEmbedder (ArcFace) + FaceRegistry
AVService
← av.say / beep / utterance / announce_version
→ av.spoke / av.speaking_started / version_announced
NotificationService
← thermal.temp / perception.faces
← av.speaking_started / av.spoke
→ av.say (thermal alerts, absence alerts)
rate-limited · quiet-hours aware
VoiceCommandService
← audio stream/VAD
→ voice.state / wake / transcript / intent
→ av.utterance (routed transcript)
wake-window + dialog context
ObjectService
← vision.frame_ready / object.set_enabled
→ object.detections
2fps · max_objects cap · BGR→RGB fix
MotionService
← motion.pan_to / relax / stop
→ motion.position / moved
TrackingService
← perception.faces
→ motion.pan_to
← av.speaking_started / av.spoke
20Hz spring-damper tracking
idle random gaze · speaking nod
ThermalService
— thermal.temp
— thermal.fan
— thermal.rpm
— thermal.critical
— thermal.error
IPCBridge (media)
PUB ipc:///tmp/desktop-assistant-media.pub
REP ipc:///tmp/desktop-assistant-media.rep
SUB ← core.pub (upstream)
RPCs: music.get_state/set_volume/...
podcast.search/subscribe/play/...
ZMQ PUB→SUB (cross-process)
ZMQ PUB→SUB (cross-process)
IPCBridge (core)
PUB ipc:///tmp/desktop-assistant.pub
REP ipc:///tmp/desktop-assistant.rep
SUB ← thermal.pub (upstream)
cmds: ping / topics / last / publish / status
PlaybackSpectrumService
← music.state_changed / podcast.playback
pw-record on sink monitor + FFT
→ display.spectrum
systemd: desktop-assistant-thermal.service
IPCBridge (thermal)
PUB ipc:///tmp/desktop-assistant-thermal.pub
REP ipc:///tmp/desktop-assistant-thermal.rep
MusicService
← music.play / stop / pause / next
← music.thumbs_up / thumbs_down / set_volume
Pandora / audio playback
PodcastService
Apple Podcasts search + RSS subscriptions
← podcast.play/pause/resume/stop/refresh
→ podcast.playback / subscriptions_updated
BLE GATT
status / mouth / eq
latest_frame()
config R/W
suppresses announcements
suppresses greetings
suppresses alerts
re-emit
extends
systemd: desktop-assistant-core.service
Arduino firmware
firmware/vera_display
(ESP32-C6 process boundary)
NimBLEOta service (BLE OTA, 0x8018)
RoomService
← room.set
→ room.updated / av.say
persists room name · visual scene
baseline histogram · sustained
divergence detection → TTS prompt
Core (src/core/)
version.py (reads /VERSION)
QuietHours
(enabled / start / end
loaded from config/quiet_hours.yaml)
core MessageBus (in-process pub/sub)
Service base (start/tick/stop)
thermal MessageBus (in-process pub/sub)
media MessageBus (in-process pub/sub)